

TransferIQ AI-Powered Soccer Player Transfer Value Prediction

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Week 1: Dataset Collection

During the first week, I focused on collecting and understanding the dataset. The primary dataset I used was from Kaggle

<https://www.kaggle.com/datasets/sanjitva/predicting-soccer-player-transfer-values>

It included attributes like player name, age, potential, overall rating, club, league, nationality, and past transfer values. To make the dataset more comprehensive, I added external data sources. I collected injury history from the Transfermarkt website, which contains detailed records of player injuries, recovery durations, and match absences. I also used the Twitter API through the Tweepy library to fetch player-related tweets to analyze public sentiment later.

The challenge I faced this week was merging data from different sources since naming conventions varied between datasets. I had to write a small preprocessing script in Python to standardize player names and remove inconsistencies.

Week 2: Data Preprocessing

In the second week, I cleaned and prepared the dataset using Python libraries such as Pandas, NumPy, and Scikit-learn. I removed missing and duplicate records using Pandas functions like `dropna` and `drop_duplicates`. I scaled numerical data using `MinMaxScaler` from Scikit-learn and label encoded categorical columns such as club, position, and nationality.

I also created new features to make the model more predictive. These included an Injury Risk Score derived from Transfermarkt data, a Form Consistency Index calculated from rolling averages of performance scores, and a Contract Duration Ratio based on remaining contract length divided by total contract length. Finally, I merged all sources into one unified dataset.

One challenge I encountered was handling missing injury data for some players. I solved this by assigning them a neutral injury score so that the model would not bias against them.

Week 3: Sentiment Analysis

In the third week, I focused on analyzing public sentiment toward players. I collected tweets mentioning footballers using the Tweepy library. I cleaned the text using regular expressions to remove URLs, emojis, and unwanted symbols. I then removed stopwords using the NLTK library and tokenized the text for analysis.

For sentiment scoring, I used the VADER Sentiment Analyzer available in NLTK. It gave polarity scores ranging from negative to positive. I computed an average sentiment score for each player and merged this into the main dataset.

The challenge I faced this week was dealing with multilingual and irrelevant tweets. I used the TextBlob library to translate non-English tweets and filtered out non-football-related content using keyword matching.

Week 4: Feature Engineering

In the fourth week, I performed advanced feature engineering to improve the quality of inputs to the model. I used Pandas to calculate statistical features such as mean and standard deviation of player performance metrics. I also analyzed correlations among the attributes using Seaborn heatmaps to identify redundant features and remove them.

At this stage, the dataset became more balanced and informative. I saved the final cleaned and processed dataset as a CSV file to be used for model training in the upcoming weeks.

The main challenge I faced during this stage was feature imbalance. Some players had too many data points while others had very few. I used sampling and normalization techniques to maintain uniformity.

Week 5: Model Development

During the fifth week, I started developing the machine learning models. I used TensorFlow and Keras for building LSTM models and XGBoost for the ensemble component.

I first developed a univariate LSTM model that focused only on the historical transfer values to capture temporal dependencies. Later, I extended it into a multivariate LSTM model by including additional features such as performance statistics, injury risk score, and sentiment score. After training both models, I used XGBoost to build another model that captured complex relationships between structured features. Finally, I combined the two models using a weighted ensemble approach to improve prediction accuracy.

The challenge I faced during model development was shape mismatch errors when feeding time-series data into the LSTM. I fixed this by reshaping the input arrays into the correct three-dimensional format of samples, timesteps, and features.

Week 6: Forecasting and Evaluation

In the sixth week, I trained the models and performed forecasting and evaluation. I used the Adam optimizer with a Mean Squared Error loss function for the LSTM model. The models were trained for 50 epochs with early stopping to avoid overfitting.

For evaluation, I used RMSE, MAE, and R^2 score as performance metrics. The ensemble model gave the lowest RMSE and produced consistent results across test data. I plotted the training and validation loss curves using Matplotlib and Seaborn to visualize how the models converged over epochs.

One of the main challenges I faced this week was overfitting in the LSTM model. I resolved it by adding dropout layers and reducing the learning rate.

Week 7: Visualization and Insights

In the seventh week, I focused on visualizing the results and interpreting the findings. I created various visualizations using Matplotlib, Seaborn, and Plotly. These included player value trend graphs, correlation heatmaps, and feature importance plots. I also generated scatter plots to analyze how sentiment and injury affected transfer values.

The key insights I observed were that young players with high potential and positive public sentiment showed higher predicted value growth. Injury frequency had a negative impact on market value regardless of player potential. Sentiment turned out to be a significant factor in short-term fluctuations, especially during transfer windows.

The challenge I faced was optimizing visual clarity when displaying multiple variables together. I overcame this by using Plotly's interactive visualization tools to make the graphs easier to interpret.

Week 8: Finalization and Reporting

In the final week, I finalized the models, evaluated results, and prepared the documentation and presentation. I verified the outputs by testing the model on unseen data samples and confirmed that the predictions aligned well with real-world market trends.

I prepared the project report summarizing the methodology, implementation, and findings. I also created visual dashboards showing the player performance trends and sentiment impact. This week helped me consolidate everything I learned over the previous weeks into a complete and presentable project.

The main challenge was combining everything into a coherent and concise format without losing technical detail. It required balancing between explanation and visual representation.

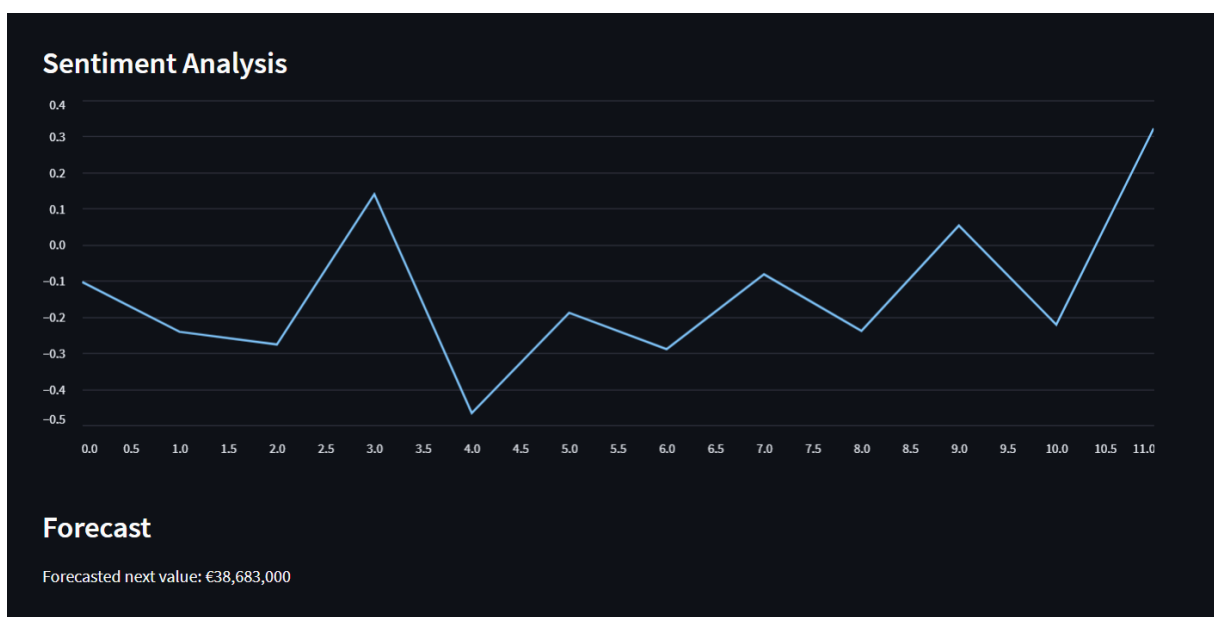
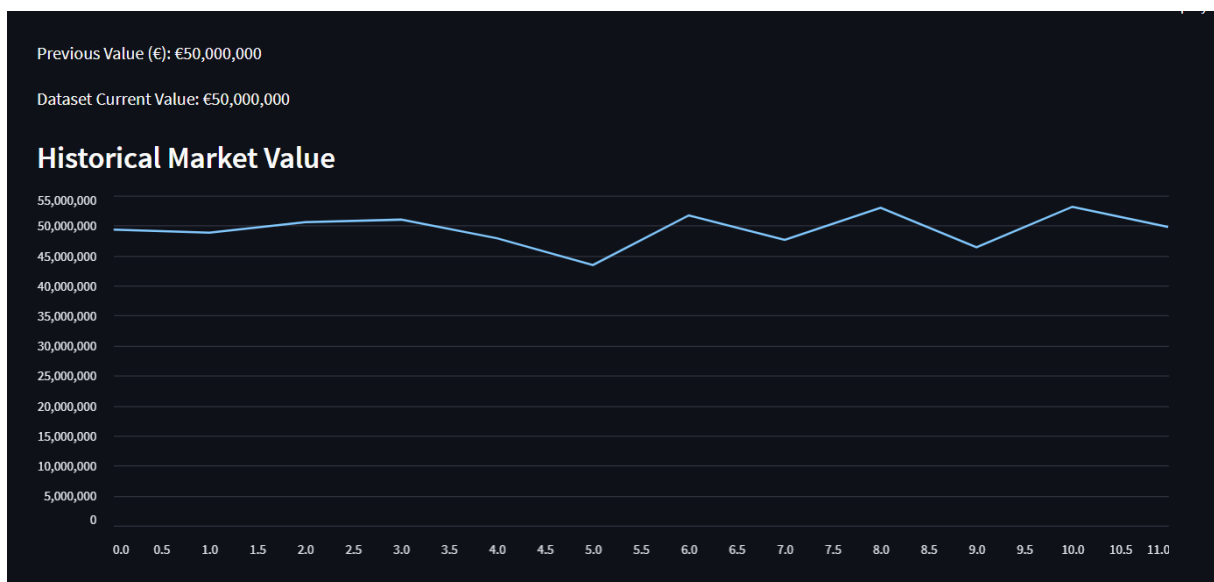
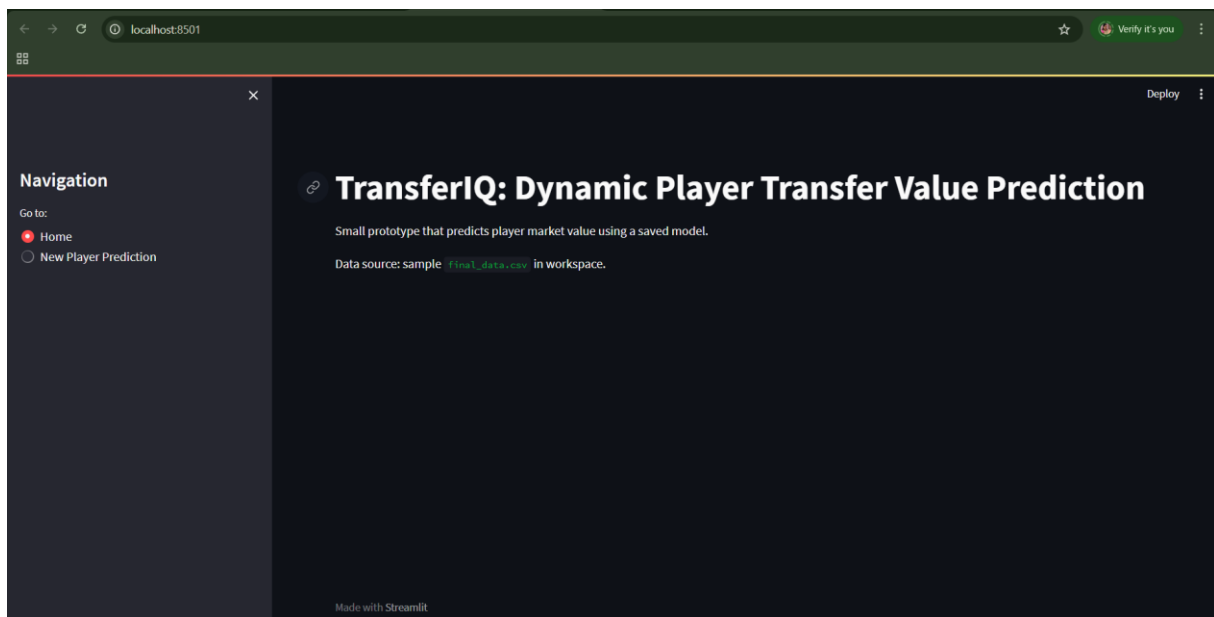
Conclusion

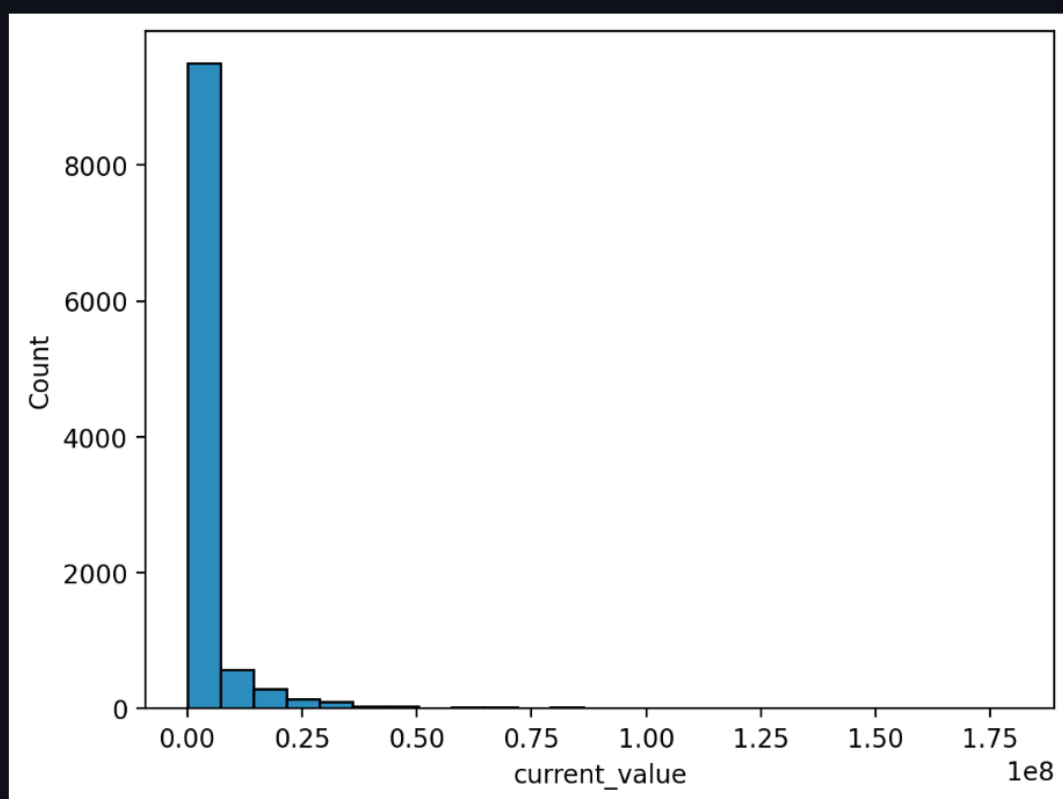
This eight-week project gave me a deep understanding of how artificial intelligence can be used to predict player transfer values using multiple data sources. I worked with data from Kaggle, Transfermarkt, and Twitter to build a predictive model using TensorFlow, XGBoost, and NLP tools.

I gained hands-on experience in data preprocessing, feature engineering, time-series forecasting, and sentiment analysis. I also learned how small changes in data quality or scaling can influence overall model performance.

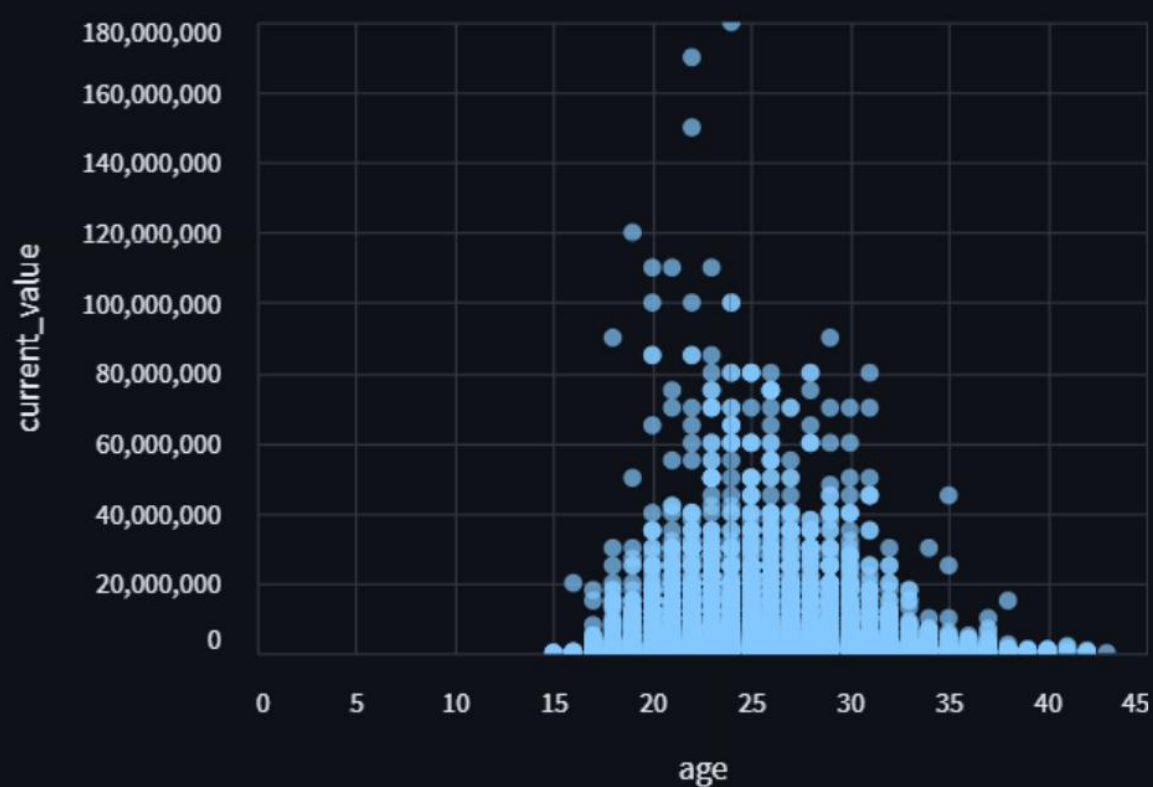
Overall, TransferIQ helped me understand the real-world application of AI and data science in sports analytics and improved my technical and analytical skills throughout the process.

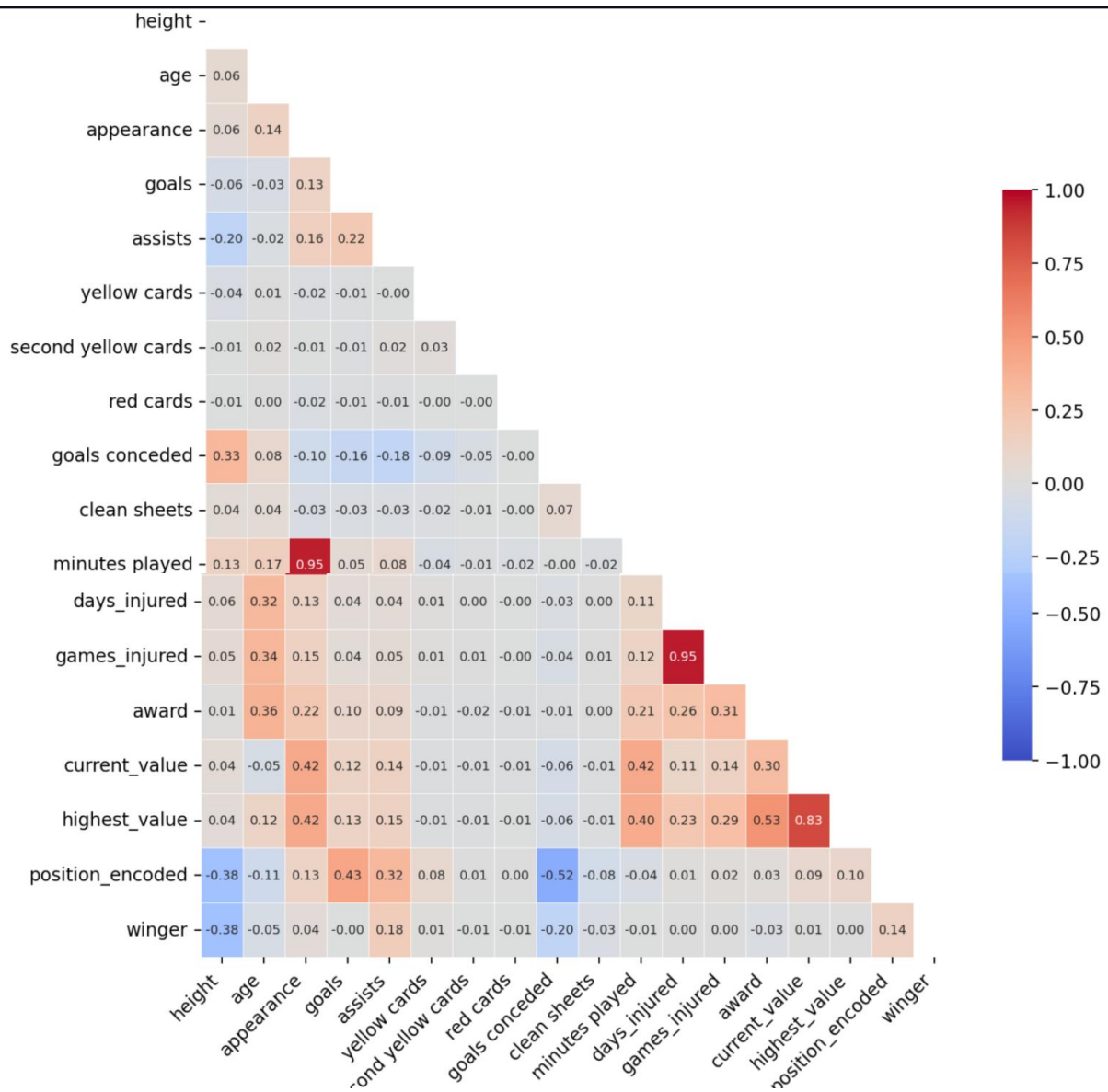
Working:





Scatter plot





Top players by current value

	name	current_value
0	Kylian Mbappé	180,000,000
1	Erling Haaland	170,000,000
2	Vinicius Junior	150,000,000
3	Jude Bellingham	120,000,000
4	Phil Foden	110,000,000
5	Jamal Musiala	110,000,000
6	Bukayo Saka	110,000,000
7	Federico Valverde	100,000,000
8	Rodrygo	100,000,000
9	Victor Osimhen	100,000,000

